

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{4x^2}{x^2-9} - \frac{2x}{x+3} = \frac{1}{x-3}$$

What value of x satisfies the equation above?

Answer

- A. -3
- B. $-\frac{1}{2}$
- C. $\frac{1}{2}$
- D. 3

Correct Answer: C

Rationale

Choice C is correct. Each fraction in the given equation can be expressed with the common denominator x^2-9 . Multiplying $\frac{2x}{x+3}$ by $\frac{x-3}{x-3}$ yields $\frac{2x^2-6x}{x^2-9}$, and multiplying $\frac{1}{x-3}$ by $\frac{x+3}{x+3}$ yields $\frac{x+3}{x^2-9}$. Therefore, the given equation can be written as $\frac{4x^2}{x^2-9} - \frac{2x^2-6x}{x^2-9} = \frac{x+3}{x^2-9}$. Multiplying each fraction by the denominator results in the equation $4x^2 - (2x^2 - 6x) = x + 3$, or $2x^2 + 6x = x + 3$. This equation can be solved by setting a quadratic expression equal to 0, then solving for x. Subtracting $x + 3$ from both sides of this equation yields $2x^2 + 5x - 3 = 0$. The expression $2x^2 + 5x - 3$ can be factored, resulting in the equation $(2x - 1)(x + 3) = 0$. By the zero product property, $2x - 1 = 0$ or $x + 3 = 0$. To solve for x in $2x - 1 = 0$, 1 can be added to both sides of the equation, resulting in $2x = 1$. Dividing both sides of this equation by 2 results in $x = \frac{1}{2}$. Solving for x in $x + 3 = 0$ yields $x = -3$. However, this value of x would result in the second fraction of the original equation having a denominator of 0. Therefore, $x = -3$ is an extraneous solution. Thus, the only value of x that satisfies the given equation is $x = \frac{1}{2}$.

Choice A is incorrect and may result from solving $x + 3 = 0$ but not realizing that this solution is extraneous because it would result in a denominator of 0 in the second fraction. Choice B is incorrect and may result from a sign error when solving $2x - 1 = 0$ for x. Choice D is incorrect and may result from a calculation error.

